

RESEARCH PROFILE

Seeking a funded Doctoral Researcher or Research Assistant position in human-computer interaction, human-AI interaction, immersive systems, mixed reality, or user research, with availability from August–September 2027.

Human-computer interaction researcher combining empirical user research with hands-on development across immersive systems, multimodal AI, machine learning, and real-time computer graphics. Research examines how proactive and relational AI shapes users' metacognition, agency, perceived control, and social acceptability, particularly in AR smart glasses. Designs and implements the prototypes and experimental apparatus used in her own studies, translating technical concepts into working systems and evaluating their effects through controlled experiments and quantitative modelling.

RESEARCH AND PROFESSIONAL EXPERIENCE

AXIS Lab, Interaction Science Institute, Sungkyunkwan University 2024–present
Undergraduate Research Assistant (2024–Feb 2025); Master's Student Researcher (Mar 2025–Aug 2026); Researcher (Sept 2026–present). Contributing to projects on proactive AI and human cognitive mechanisms in always-on AI environments, and continuing prototype-based research on AR smart glasses, socially visible sensing, and immersive human-AI interaction.

Researcher and App Developer, Personalized AI Story Generation Project Apr 2026 – present
Developing an application for AI-generated personalized children's stories based on everyday-object recognition and developmental stage; project supported by Seoul RISE and led by Prof. Changjun Lee.

Assistant Instructor, Generative AI Content Creation Workshop, Hyundai Futurenet Jul 2026
Supported hands-on workshop activities on generative-AI content creation, including participant guidance and practical exercises for producing AI-assisted visual and interactive media.

Research Intern, RGB Makers Co., Ltd. Aug 2025
Analysed first-person interaction data and privacy risks in metahuman-based content, developed a MetaHuman character in Unreal Engine 5, and proposed outdoor interaction scenarios for user-centred immersive experiences.

BK21 Research Assistant, Social Interaction Big Data Education and Research Group Mar – Jun 2025
Supported immersive-device user research at AXIS Lab through literature review, study and protocol preparation, participant recruitment materials, lab setup, research ethics documentation, and data management.

EDUCATION

Sungkyunkwan University · Master of Science in Engineering, Immersive Media Engineering Mar 2025 – Aug 2026
Advisor: Prof. Changjun Lee · AXIS Lab · Cumulative GPA: 4.45/4.50 (99.5/100)

Thesis: *When Sensing Becomes Socially Visible: Wearer Meaning and Adoption Resistance in AR Smart Glasses*

Sungkyunkwan University · Bachelor of Culture & Technology; Bachelor of Economics Feb 2021 – Feb 2025
Major in Culture and Technology Convergence; International Trade Linked Major, Economics Track. Cumulative GPA: 4.32/4.50 (98.2/100). Graduated with SKKU Summa Cum Laude.

PUBLICATIONS

PEER-REVIEWED CONFERENCE PUBLICATIONS

Kim, G., Choi, Y., Kim, Y., & Lee, C. (2026). No Time to Say Goodbye: Emotional Loss Responses to Sudden Termination in Immersive AI Interactions. *CHI EA '26: Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. ACM.

Kim, G., Choi, Y., & Lee, C. (2025). See, Sense, Spark: Profiling Granularity in Proactive AI for Creative Research Ideation. SIGGRAPH Asia 2025 Posters. ACM. DOI: 10.1145/3757374.3771458

Kim, G., & Han, Y. (2024). Camp_Around: An Integrated Tour Service for International Students in Korea. *HCI Korea 2024 Creative Award Proceedings*, 1224–1227.

JOURNAL ARTICLES

Kim, G., Kim, S. Y., Park, S., Park, S., Choi, Y., & Lee, C. (2025). What Makes Audiences Pay More for Immersive Cinema? A Conjoint Analysis of ScreenX Preferences and Willingness-to-Pay. *International Telecommunications Policy Review*, 32(4), 1–45. DOI: 10.37793/ITPR.32.4.1

Kim, G., Choi, Y., & Lee, C. (2025). The Impact of Pre-adulthood Immersive Media Experiences on Continued Use Intention in Adulthood: An Empirical Study Applying the Value-based Adoption Model and Positive Technology Readiness Index. *Information Society & Media*, 26(1), 143–183.

MANUSCRIPTS UNDER REVIEW

Journal manuscript under review. Socially situated sensing and technology adoption in AR smart glasses.

Two manuscripts under review at CHI 2027. Human–AI interaction, proactive interfaces, and immersive relational systems.

SELECTED FUNDED RESEARCH PROJECTS

Youth-Focused AI Risk Evaluation Benchmark Development	Researcher · ETRI · 2026
Proactive AI and Human Cognitive Mechanisms in Always-On AI Environments	Researcher · NRF · 2026–present
Personalized AI Story Generation for Children	Researcher · Seoul RISE · 2026–present
Interactive Experiential News Game Research and Development	Researcher · SKKU RISE · 2025–2026
Screen Multimodal Interaction Research · Big Data Analytics Team	Researcher · Samsung Electronics · 2024
Virtual-World Multi-Persona Tendencies and Cognitive Enhancement	Researcher · NRF Korea · 2025
Urban Green Exposure, Mobility, and Physical and Mental Health · Researcher · AI Convergence Research Support · 2025	
Humanities–Engineering Convergence Research on Psychological Counselling Chatbots · Researcher · NRF Korea · 2026	
News and Social-Media Issue Analysis for the National Health Insurance Service	Researcher · NHIS · 2024

SELECTED RESEARCH AND PROJECTS

AR Smart Glasses and Socially Visible Sensing	Master's thesis, 2026
– Examined how sensing that is visible to bystanders, not only to the wearer, shapes wearer meaning and adoption resistance in AR smart glasses.	
– Third study in a programme of work conducted on AR smart glasses hardware, alongside the proactive-AI and AI-companion experiments below.	

See, Sense, Spark — Proactive AI for Creative Research Ideation	SIGGRAPH Asia 2025
– Built a proactive AI system on Xreal Air 2 Pro smart glasses and ran a between-subjects study of profiling granularity; fine-grained profiles increased interest-driven curiosity and affective engagement.	

No Time to Say Goodbye — Termination and Emotional Loss in AI Companions	CHI EA 2026
– Built IRO, an immersive AI companion, and conducted a between-subjects experiment (N = 26); advance forewarning reduced loss responses (ANCOVA, $p = .011$), supporting relational offboarding as a distinct UX phase.	

Immersive Media Adoption and Value Perception	2025 journal studies
– Studied immersive-media adoption, perceived value, and willingness-to-pay using structural equation modelling, discrete-choice conjoint analysis, and user segmentation.	

Phodong: AI-Based Phygital Story Creation for Children	HCI Korea 2026
– Developed a generative AI and object-recognition concept for parent–child media engagement and child-centred storytelling.	

Smart Farming, Bright Future · Game-Based Learning Simulation	Researcher · UNESCO GHE · 2025
– Co-developed an educational smart-farming simulation with ten science-based missions using real-time environmental data manipulation.	

SELECTED TECHNICAL DEVELOPMENT

3D Gaussian Splatting Reconstruction and Viewer	WebGL / GLSL, 2025
– Built a COLMAP and 3D Gaussian Splatting reconstruction workflow and reimplemented core forward-rasterization, alpha blending, depth sorting, and camera-aware optimization in JavaScript and GLSL.	

Neural Models for Visual and Text Classification	PyTorch, 2026
– Designed and evaluated an MLP for Fashion-MNIST and a padding-aware bidirectional LSTM for IMDb sentiment classification, with controlled comparisons of augmentation, regularization, pooling, and training configurations.	

Interactive Graphics and Geometry Processing	Three.js / WebGL / GLSL, 2025
– Implemented articulated animation and interactive graphics in Three.js, plus half-edge mesh processing, Catmull–Clark subdivision, and derivative-based texture filtering in GLSL.	

AWARDS, EXHIBITIONS, AND CREATIVE WORK

HCI Korea 2026 Creative Award · Grand Prize	2026
Selected Team, 2026 Promising Student Startup Team 300+ · Growth Track	2026
GS SHOP Content Leader Scholarship · 13th Cohort	Jan 2025
SKKU Summa Cum Laude · Sungkyunkwan University	Feb 2025
Best Award · 2nd Joint Academic Festival of Convergence Colleges	Aug 2024
Best Presentation Award · Cybercommunication Society Spring Conference	May 2024
Ars Electronica 2024 · Exhibited work, Linz	2024
HCI Korea 2024 Creative Award Proceedings · Published contribution	2024
Grand Prize · 2nd Culture and Technology Convergence Major Day Capstone Project Competition	Dec 2023

LEADERSHIP AND ACADEMIC SERVICE

President, 4th Student Council CNT:role, Culture and Technology Convergence	2023
Led student community programmes at Sungkyunkwan University and helped launch the first joint academic festival across five convergence colleges.	

RESEARCH SKILLS

Programming and data	Python; R; JavaScript; NumPy; pandas
AI, XR, and graphics	PyTorch; COLMAP; Unity; Unreal Engine; Three.js; WebGL; GLSL — used for neural models, 3D reconstruction, real-time graphics, immersive prototypes, and experimental systems
Research methods	Controlled experiments; between-subjects designs; survey experiments; structural equation modelling; discrete-choice conjoint analysis; willingness-to-pay estimation; clustering and segmentation; prototype-based user studies
Research areas	Human–AI interaction; user agency and metacognition; proactive and creative AI; AR smart glasses and wearable XR; AI companions; technology adoption
Languages	Korean (native); Chinese (proficient — six years of residence in China); English (academic reading, writing, and research communication)

REFERENCES

Prof. Changjun Lee · M.S. advisor	Sungkyunkwan University
AXIS Lab, Department of Immersive Media Engineering · changjunlee@skku.edu	